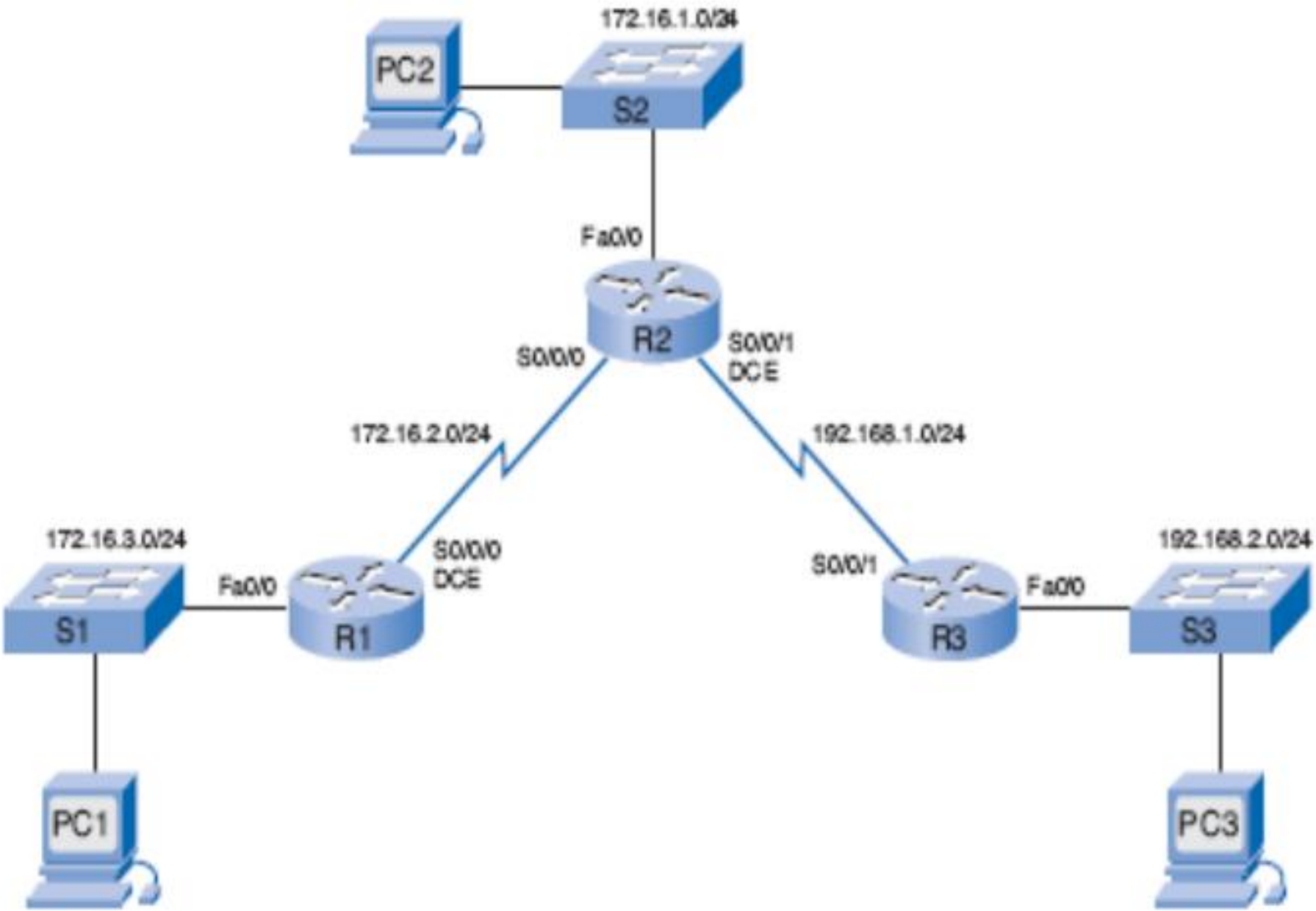
Static and dynamic route configuration

Chapter Three

Static Routing

- **Remote networks** are added to the routing table either by configuring static routes or enabling a dynamic routing protocol.
- A static route includes the **network address and subnet mask of the remote network, along with the IP address of the next-hop router or exit interface.**
- Static routes should be used in the following cases:
  - **A network consists of only a few routers.**
  - **A network is connected to the Internet only through a single ISP.**
  - **A large network is configured in a hub-and-spoke topology.**

Introducing the Topology



Interfaces and their Statuses

```
R1# show interfaces
FastEthernet0/0 is administratively down, line protocol is down
Hardware is AmdFE, address is 000c.3010.9260 (bia
   000c.3010.9260)
<output omitted>
Serial0/0/0 is administratively down, line protocol is down
<output omitted>
```

```
R1# show ip interface brief
Interface        IP-Address OK? Method Status                Protocol
FastEthernet0/0 unassigned YES manual administratively down down
Serial0/0       unassigned YES unset  administratively down down
FastEthernet0/1 unassigned YES unset  administratively down down
Serial0/1       unassigned YES unset  administratively down down
R1#
```

- **Used to see a portion of the interface information in a condensed format**

Configuring an Ethernet Interface

```
R1(config)# interface fastethernet 0/0
R1(config-if)# ip address 172.16.3.1 255.255.255.0
R1(config-if)# no shutdown
*Mar 1 01:16:08.212: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 01:16:09.214: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/0, changed state to up
```

- **changed state to up** message indicates that, **physically, the connection is good.**
  - If you do not get this first message, be sure that the interface is properly connected to a switch or a hub.
  - An Ethernet interface will not be active, or up, **unless it is receiving a carrier signal from another device** (switch, hub, PC, or another router).
- **changed state to up** message indicates that the data link layer is operational.
  - On LAN interfaces, you **do not** normally change the data link layer parameters.
  - However, WAN interfaces in a lab environment require clocking on one side of the link.

Commands to Verify Interface Configuration

```
R1# show interfaces fastethernet 0/0
FastEthernet0/0 is up, line protocol is up
Hardware is AmdFE, address is 000c.3010.9260 (bia 000c.3010.9260)
Internet address is 172.16.3.1/24
<output omitted>

R1# show ip interface brief
Interface        IP-Address OK? Method Status              Protocol
FastEthernet0/0 172.16.3.1 YES manual up                   up
Serial0/0/0      unassigned YES unset  administratively down down
FastEthernet0/1 unassigned YES unset  administratively down down
Serial0/0/1      unassigned YES unset  administratively down down

R1# show running-config
<output omitted>
interface FastEthernet0/0
ip address 172.16.3.1 255.255.255.0
<output omitted>
```

Configuring a Serial Interface

```
R1(config)# interface serial 0/0/0
R1(config-if)# ip address 172.16.2.1 255.255.255.0
R1(config-if)# no shutdown

R1# show interfaces serial 0/0/0
Serial0/0/0 is down, line protocol is down
Hardware is PowerQUICC Serial
Internet address is 172.16.2.1/24
MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
<output omitted>
```

- After entering these commands, the state of the serial interface might vary depending on the type of WAN connection.
- The serial interface will be in the up state **only after the other end of the serial link has also been properly configured.**

Configuring a Serial Interface

```
R1(config)# interface serial 0/0/0
R1(config-if)# ip address 172.16.2.1 255.255.255.0
R1(config-if)# no shutdown
```

```
R2(config)# interface serial 0/0/0
R2(config-if)# ip address 172.16.2.2 255.255.255.0
R2(config-if)# no shutdown
```

- There is no requirement that both ends of the serial link use the same interface, in this case, Serial 0/0/0.
- However, because both interfaces are members of the same network, they both must have IP addresses that **belong to the 172.16.2.0/24 network.**

Configuring a Serial Interface

```
R2# show interfaces serial 0/0/0
Serial0/0/0 is up, line protocol is down
<output omitted>
```

- Physical link between R1 and R2 is up because both ends of the serial link have been configured correctly with an IP address/mask and enabled with the **`no shutdown`** command.
- However, the line protocol is still down.
- This is because the interface is not **receiving a clock signal.**
- There is still one more command that you need to enter, the **`clock rate`** command, on the router with the **DCE cable.**
- The **`clock rate`** command will set the clock signal for the link.
- Typically, the router is the DTE device and is connected to a CSU/DSU, which is the DCE device.
- **Serial interfaces require a clock signal to control the timing of the communications.**
- In most environments, the service provider (a DCE device such as a CSU/DSU) will provide the clock.
- By default, routers are DTE devices.

Configuring Serial Links in a Lab Environment

- In a lab environment, one side of a connection must be considered a **DCE and provide a clocking signal.**
- Although serial interfaces are DTE devices by default, they can be configured as DCE devices.

```
R1(config)# interface serial 0/0/0
R1(config-if)# clock rate 64000
01:10:28: %LINEPROTO-5-UPDOWN: Line protocol on
  Interface Serial0/0/0, changed state to up
```

- Available clock rates, in bits per second, are 1200, 2400, 9600, 19200, 38400, 56000, 64000, 72000, 125000, 148000, 500000, 800000, 1000000, 1300000, 2000000, and 4000000.
- If a router's interface with a DTE cable is configured with the **clock rate** command, the IOS will disregard the command and there will be no ill effects.

Verifying the Serial Interface Configuration

```
R1# show interfaces serial 0/0/0
Serial0/0/0 is up, line protocol is up
Hardware is PowerQUICC Serial
Internet address is 172.16.2.1/24
<output omitted>

R1# show ip interface brief
Interface          IP-Address OK? Method   Status Protocol
FastEthernet0/0 172.16.3.1 YES manual   up     up
Serial0/0/0        172.16.2.1 YES manual   up     up
<output omitted>
```

```
R1# show ip route
<output omitted>
Gateway of last resort is not set
     172.16.0.0/24 is subnetted, 2 subnets
C       172.16.2.0 is directly connected, Serial0/0/0
C       172.16.3.0 is directly connected, FastEthernet0/0
```

- 172.16.2.0/24 serial network is now in the routing table for R1

Routing Table Concepts

- The routing table consists of a list of "known" **network** addresses—that is, those addresses that are **directly** connected, **configured** statically, and **learned** dynamically.
- R1 and R2 only have routes for directly connected networks.

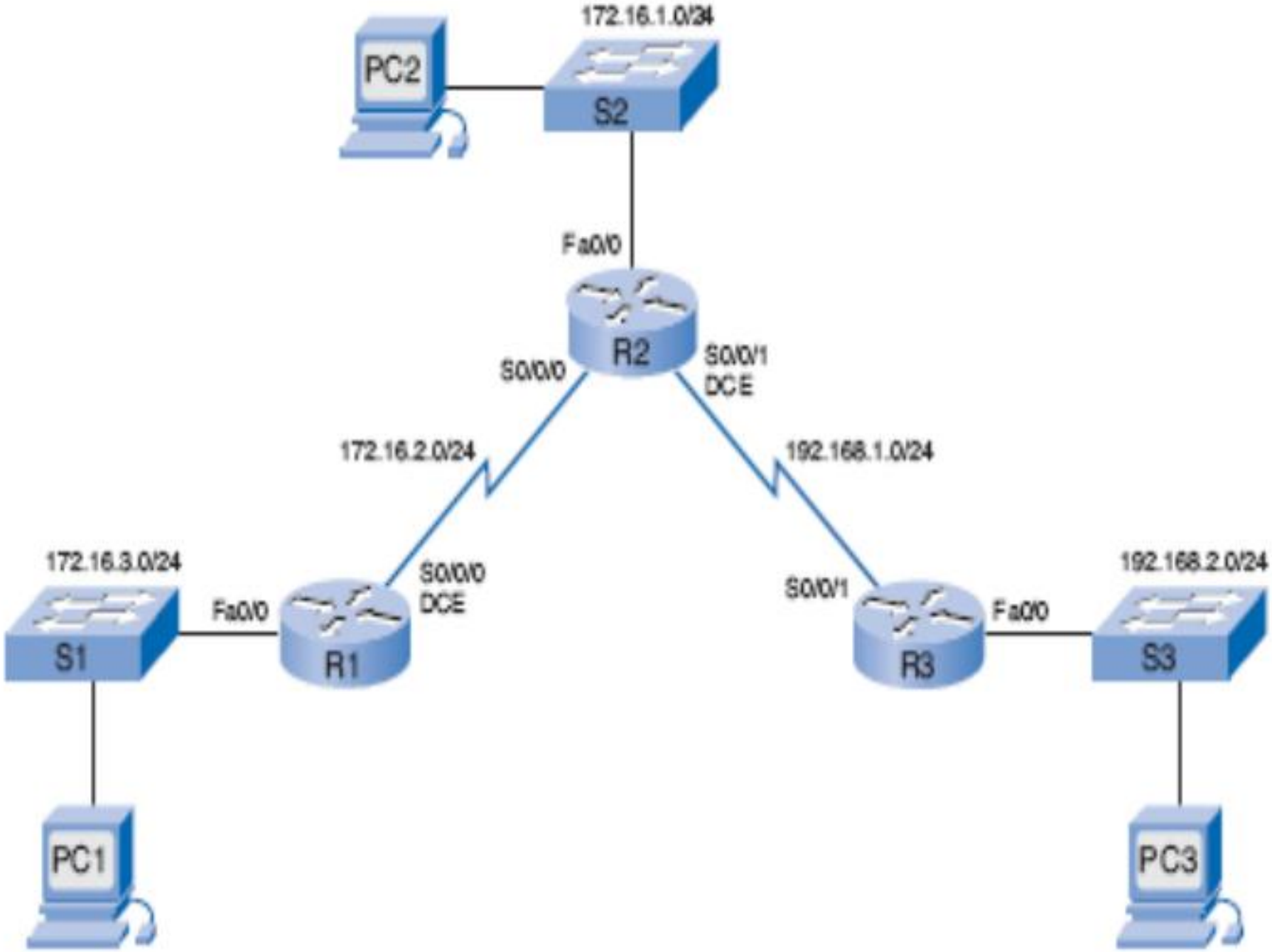
```
R1# show ip route
<output omitted>
Gateway of last resort is not set
     172.16.0.0/24 is subnetted, 2 subnets
C       172.16.2.0 is directly connected, Serial0/0/0
C       172.16.3.0 is directly connected, FastEthernet0/0
```

```
R2# show ip route
<output omitted>
Gateway of last resort is not set
     172.16.0.0/24 is subnetted, 1 subnets
C       172.16.2.0 is directly connected, Serial0/0/0
```

The rest of the configurations for Routers R2 and R3.

```
R2(config)# interface serial 0/0/1
R2(config-if)# ip address 192.168.1.2 255.255.255.0
R2(config-if)# clock rate 64000
R2(config-if)# no shutdown
R2(config-if)# interface fastethernet 0/0
R2(config-if)# ip address 172.16.1.1 255.255.255.0
R2(config-if)# no shutdown
```

```
R3(config)# interface fastethernet 0/0
R3(config-if)# ip address 192.168.2.1 255.255.255.0
R3(config-if)# no shutdown
R3(config-if)# interface serial 0/0/1
R3(config-if)# ip address 192.168.1.1 255.255.255.0
R3(config-if)# no shutdown
```


172.16.1.0/24
PC2
S2
Fa0/0
R2
S0/0/0
S0/0/1
DCE
172.16.2.0/24
192.168.1.0/24
172.16.3.0/24
S0/0/0
DCE
S0/0/1
192.168.2.0/24
Fa0/0
Fa0/0
S1
R1
R3
S3
PC1
PC3

Verifying Configurations

```
R1# show ip route
     172.16.0.0/24 is subnetted, 2 subnets
C       172.16.2.0 is directly connected, Serial0/0/0
C       172.16.3.0 is directly connected, FastEthernet0/0
```

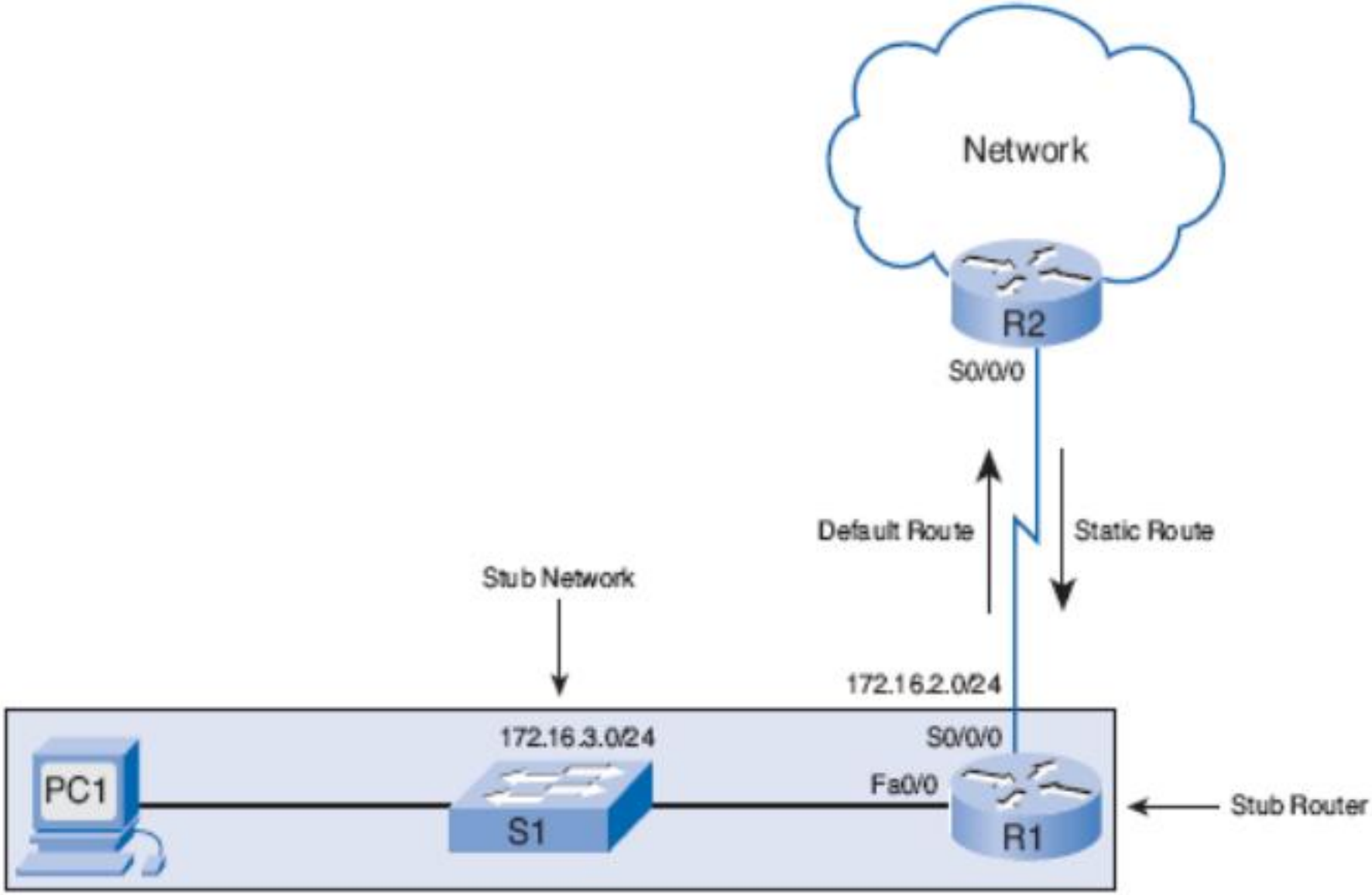
```
R2# show ip route

     172.16.0.0/24 is subnetted, 2 subnets
C       172.16.1.0 is directly connected, FastEthernet0/0
C       172.16.2.0 is directly connected, Serial0/0/0
C       192.168.1.0/24 is directly connected, Serial0/0/1
```

```
R3# show ip route

C    192.168.1.0/24 is directly connected, Serial0/0/1
C    192.168.2.0/24 is directly connected, FastEthernet0/0
```

Static Routes with "Next-Hop" Addresses



Static routes

- Static routes are commonly used when **routing from a network to a stub network.**
- A ***stub network*** is a network **accessed by a single route.**
- Here you see that any network attached to R1 would only have one way to reach other destinations, whether to networks attached to R2 or to destinations beyond R2.
- Therefore, network **172.16.3.0 is a stub network** and R1 is a ***stub router***.
- Running a routing protocol between R1 and R2 is a waste of resources because R1 has only one way out for sending nonlocal traffic.
- Therefore, static routes are configured for connectivity to **remote** networks that are **not directly connected** to a router.

ip route Command

The command for configuring a static route is **ip route**.

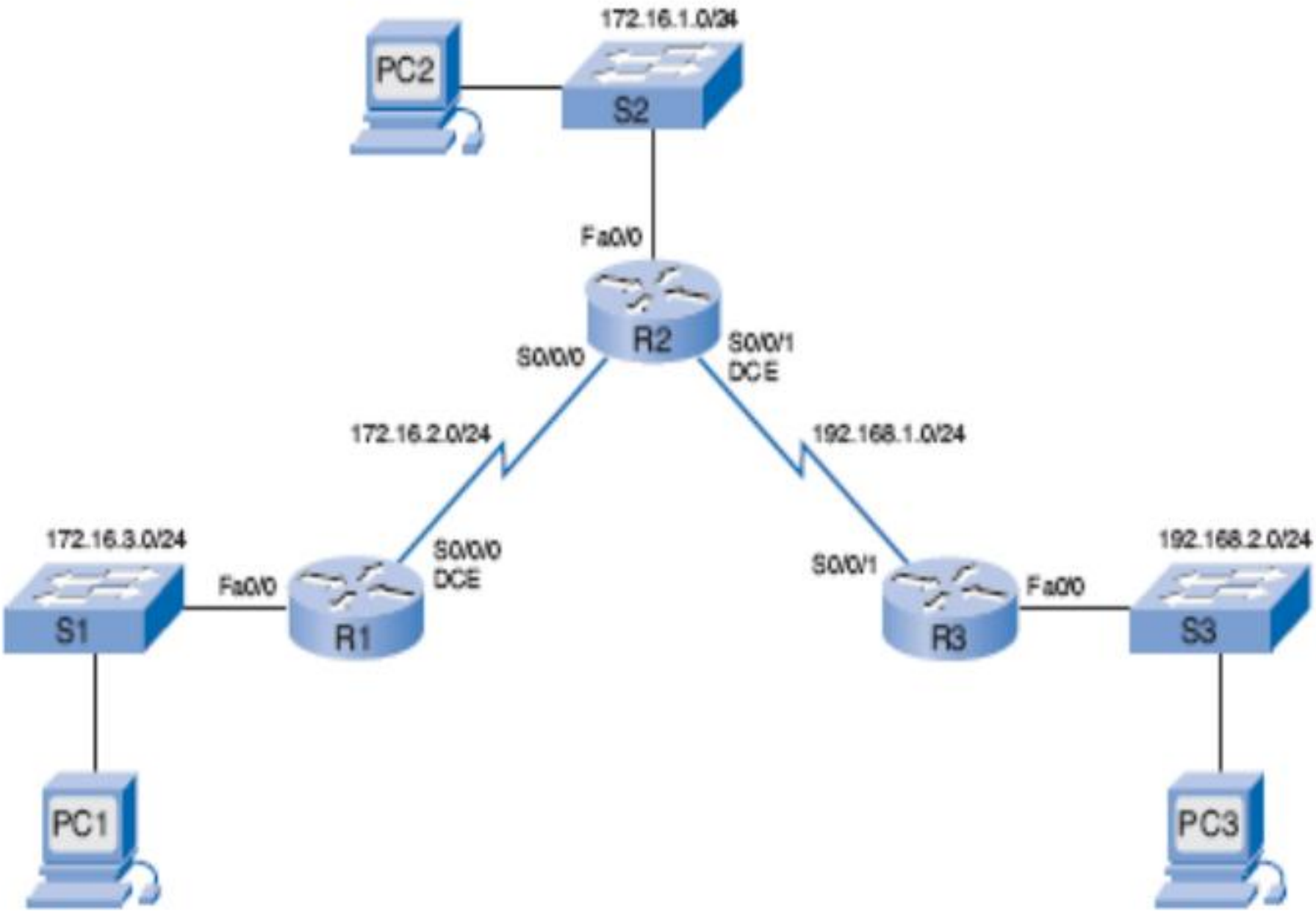
Simpler version of the syntax:

```
Router(config)# ip route network-address subnet-mask
  {ip-address | exit-interface}
```

- The following parameters are used:
  - ***network-address***: Destination network address of the remote network to be added to the routing table.
  - ***subnet-mask*:** Subnet mask of the remote network to be added to the routing table. The subnet mask can be modified to summarize a group of networks.

- One or both of the following parameters must also be used:
  - ***ip-address*:** Commonly referred to as the **next-hop router's IP address.**
  - ***exit-interface*:** Outgoing interface that would be used in forwarding packets to the destination network.

Introducing the Topology



Configuring Static Routes

- R1 in our topology knows about its directly connected networks.
- The remote networks that R1 **does not know** about are as follows:
  - **172.16.1.0/24:** The LAN on R2
  - **192.168.1.0/24:** The serial network between R2 and R3
  - **192.168.2.0/24:** The LAN on R3

```
R1(config)# ip route 172.16.1.0 255.255.255.0 172.16.2.2
```

- Examining each element in this output reveals the following:
  - **`ip route`:** Static route command
  - **`172.16.1.0`:** Network address of remote network
  - **`255.255.255.0`:** Subnet mask of remote network
  - **`172.16.2.2`:** Serial 0/0/0 interface IP address on **R2,** which is the **next hop to this network.**
- When the IP address is the actual next-hop router's IP address, this IP address is reachable from one of this router's directly connected networks.
- In other words, the next-hop IP address 172.16.2.2 belongs to Router R1's directly connected Serial 0/0/0 network 172.16.2.0/24..

Configuring Routes to Two or More Remote Networks

```
R1(config)# ip route 192.168.1.0 255.255.255.0 172.16.2.2
R1(config)# ip route 192.168.2.0 255.255.255.0 172.16.2.2
R1(config)# end
R1# show ip route
     172.16.0.0/24 is subnetted, 3 subnets
S       172.16.1.0 [1/0] via 172.16.2.2
C       172.16.2.0 is directly connected, Serial0/0/0
C       172.16.3.0 is directly connected, FastEthernet0/0
S 192.168.1.0/24 [1/0] via 172.16.2.2
S 192.168.2.0/24 [1/0] via 172.16.2.2
```

- Notice that all three static routes configured on R1 have the same next-hop IP address: 172.16.2.2, because packets for all the remote networks must be forwarded to Router R2, the next hop router.

Configure R2 and R3 Static Routes

```
R2(config)# ip route 172.16.3.0 255.255.255.0 172.16.2.1
R2(config)# ip route 192.168.2.0 255.255.255.0 192.168.1.1
```

```
R3(config)# ip route 172.16.1.0 255.255.255.0 192.168.1.2
R3(config)# ip route 172.16.2.0 255.255.255.0 192.168.1.2
R3(config)# ip route 172.16.3.0 255.255.255.0 192.168.1.2
```

```
R1# show ip route
<output omitted>
     172.16.0.0/24 is subnetted, 3 subnets
S       172.16.1.0 [1/0] via 172.16.2.2
C       172.16.2.0 is directly connected, Serial0/0/0
C       172.16.3.0 is directly connected, FastEthernet0/0
S    192.168.1.0/24 [1/0] via 172.16.2.2
S    192.168.2.0/24 [1/0] via 172.16.2.2

R2# show ip route
<output omitted>
     172.16.0.0/24 is subnetted, 3 subnets
C       172.16.1.0 is directly connected, FastEthernet0/0
C       172.16.2.0 is directly connected, Serial0/0/0
S       172.16.3.0 [1/0] via 172.16.2.1
C    192.168.1.0/24 is directly connected, Serial0/0/1
S    192.168.2.0/24 [1/0] via 192.168.1.1

R3# show ip route
<output omitted>
     172.16.0.0/24 is subnetted, 3 subnets
S       172.16.1.0 [1/0] via 192.168.1.2
S       172.16.2.0 [1/0] via 192.168.1.2
S       172.16.3.0 [1/0] via 192.168.1.2
C    192.168.1.0/24 is directly connected, Serial0/0/1
C    192.168.2.0/24 is directly connected, FastEthernet0/0
```

Static Routes with Exit Interfaces

Router(config)# **ip route** *network-address subnet-mask* {*ip-address* | ***exit-interface***}

```
R1(config)# no ip route 192.168.2.0 255.255.255.0 172.16.2.2
R1(config)# ip route 192.168.2.0 255.255.255.0 serial 0/0/0
R1(config)# end
R1# show ip route
<output omitted>
     172.16.0.0/24 is subnetted, 3 subnets
S       172.16.1.0 [1/0] via 172.16.2.2
C       172.16.2.0 is directly connected, Serial0/0/0
C       172.16.3.0 is directly connected, FastEthernet0/0
S    192.168.1.0/24 [1/0] via 172.16.2.2
S    192.168.2.0/24 is directly connected, Serial0/0/0
```

- Notice that the entry in the routing table no longer refers to the next-hop IP address but refers directly to **the exit interface.**
- This exit interface is the same one that the static route was resolved to when it used the next-hop IP address.
- Resolve the route to an exit interface in a single lookup, the same Serial 0/0/0 interface.

Modifying Static Routes

- There is no way to modify an existing static route.
- The static route must be deleted and a new one configured.

```
R1(config)# no ip route 172.16.1.0 255.255.255.0 172.16.2.2
R1(config)# ip route 172.16.1.0 255.255.255.0 serial 0/0/0
R1(config)# no ip route 192.168.1.0 255.255.255.0 172.16.2.2
R1(config)# ip route 192.168.1.0 255.255.255.0 serial 0/0/0

R2(config)# no ip route 172.16.3.0 255.255.255.0 172.16.2.1
R2(config)# ip route 172.16.3.0 255.255.255.0 serial 0/0/0
R2(config)# no ip route 192.168.2.0 255.255.255.0 192.168.1.1
R2(config)# ip route 192.168.2.0 255.255.255.0 serial 0/0/1

R3(config)# no ip route 172.16.1.0 255.255.255.0 192.168.1.2
R3(config)# ip route 172.16.1.0 255.255.255.0 serial 0/0/1
R3(config)# no ip route 172.16.2.0 255.255.255.0 192.168.1.2
R3(config)# ip route 172.16.2.0 255.255.255.0 serial 0/0/1
R3(config)# no ip route 172.16.3.0 255.255.255.0 192.168.1.2
R3(config)# ip route 172.16.3.0 255.255.255.0 serial 0/0/1
```

Summary and Default Static Routes

•Summarizing Routes to Reduce the Size of the Routing Table

- ***Summary route*** is a single route that can be used to represent multiple routes.
- Summary routes are generally a set of contiguous networks (but do not have to be) that have the same exit interface or next hop IP address.
- Example:
  - 10.0.0.0/16, 10.1.0.0/16, 10.2.0.0/16, 10.3.0.0/16, 10.4.0.0/16, 10.5.0.0/16, all the way through 10.255.0.0/16
- Can be represented by a single network address: **10.0.0.0/8.**
- Creating smaller routing tables makes the routing table lookup process more efficient, because there are fewer routes to search.
- If one static route can be used instead of multiple static routes, the size of the routing table will be reduced.
- In many cases, a single static route can be used to represent **dozens, hundreds, or even thousands of routes.**

Summarizing Routes to Reduce the Size of the Routing Table

```
R3:
ip route 172.16.1.0 255.255.255.0 Serial0/0/1
ip route 172.16.2.0 255.255.255.0 Serial0/0/1
ip route 172.16.3.0 255.255.255.0 Serial0/0/1
```

- All three routes are forwarding traffic out the same Serial 0/0/1 interface can be summarized to the single 172.16.0.0 255.255.252.0

```
R3(config)# no ip route 172.16.1.0 255.255.255.0 serial0/0/1
R3(config)# no ip route 172.16.2.0 255.255.255.0 serial0/0/1
R3(config)# no ip route 172.16.3.0 255.255.255.0 serial0/0/1

R3(config)# ip route 172.16.0.0 255.255.252.0 serial0/0/1
```

- Delete individual static routes.
- Configure single summary route.

Default Static Route

- A default route is used to represent all routes **with zero or no bits matching.**
- In other words, when there are no routes that have a more specific match, the default route will be a match.
- A default static route is a route that will match **all packets.**
- Default static routes are used:
  - To represent destination networks outside the router's own routing domain.
    - A common use is when connecting a company's edge router to the ISP network.
  - **When no other routes in the routing table match** the packet's destination IP address—in other words, **when a more specific match does not exist.**
  - When a router **has only one other router** to which it is connected.
  - This condition is known as a *stub router*.

Configuring a Default Static Route

```
Router(config)# ip route 0.0.0.0 0.0.0.0 [exit-interface
   | ip-address ]
```

- 0.0.0.0 0.0.0.0 network address and mask is called a ***quad-zero route***.

```
R1(config)# no ip route 172.16.1.0 255.255.255.0 serial
   0/0/0
R1(config)# no ip route 192.168.1.0 255.255.255.0
   serial 0/0/0
R1(config)# no ip route 192.168.2.0 255.255.255.0
   serial 0/0/0
R1(config)# ip route 0.0.0.0 0.0.0.0 serial 0/0/0
```

- Delete individual static routes.
- Configure single default static route.

Verifying a Default Static Route

```
R1# show ip route
<some codes omitted>
* - candidate default, U - per-user static route, o - ODR

Gateway of last resort is 0.0.0.0 to network 0.0.0.0
     172.16.0.0/24 is subnetted, 2 subnets
C       172.16.2.0 is directly connected, Serial0/0/0
C       172.16.3.0 is directly connected, FastEthernet0/0
S*    0.0.0.0/0 is directly connected, Serial0/0/0
```

- A /0 mask indicates that **zero or no bits are needed to match.**
- As long as a more specific match doesn't exist, the default static route will match all packets.
- Default routes are very common on routers.
- Instead of routers having to store routes for all the networks in the Internet, they can store a single default route to represent any network that is not in the routing table.

Dynamic Routing

- Dynamic routing protocols are used by routers to share information about the reachability and status of remote networks.
- Dynamic routing protocols perform several activities, including:
  - Network discovery
  - Updating and maintaining routing tables.
- **IP Routing Protocols - S**ome of the more common dynamic routing protocols for routing IP packets:
  - **RIP (Routing Information Protocol)**
  - **IGRP (Interior Gateway Routing Protocol)**
  - **EIGRP (Enhanced Interior Gateway Routing Protocol)**
  - **OSPF (Open Shortest Path First)**
  - **IS-IS (Intermediate System-to-Intermediate System)**
  - **BGP (Border Gateway Protocol)**
- routers contain a combination of static routes and dynamic routes in the routing tables.

The Purpose of Dynamic Routing Protocols

A routing protocol is a set of processes, algorithms, and messages that are used to **exchange routing information** and populate the routing table with the routing protocol's choice of best paths.

The purpose of a dynamic routing protocol includes:

- **Discovery of remote networks**
- **Maintaining up-to-date routing information**
- **Choosing the best path to destination networks**
- **Ability to find a new best path if the current path is no longer available.**

The Purpose of Dynamic Routing Protocols

Dynamic Routing Protocol Operation

- All routing protocols have the same purpose - **to learn about remote networks and to quickly adapt whenever there is a change in the topology.**
- The operations of a dynamic routing protocol vary depending upon the type of routing protocol and the routing protocol itself.
- In general, the operations of a dynamic routing protocol can be described as follows:
  - **The router sends and receives routing messages on its interfaces.**
  - **The router shares routing messages and routing information with other routers that are using the same routing protocol.**
  - **Routers exchange routing information to learn about remote networks.**
  - **When a router detects a topology change the routing protocol can advertise this change to other routers.**

Static and dynamic routing advantages and disadvantages

- **Static routing advantages:**
  - Minimal CPU processing.
  - Easier for administrator to understand.
  - Easy to configure.
- **Static routing disadvantages:**
  - Configuration and maintenance is time-consuming.
  - Configuration is error-prone, especially in large networks.
  - Administrator intervention is required to maintain changing route information.
  - Does not scale well with growing networks; maintenance becomes cumbersome.
  - Requires complete knowledge of the whole network for proper implementation.

Dynamic routing advantages and disadvantages

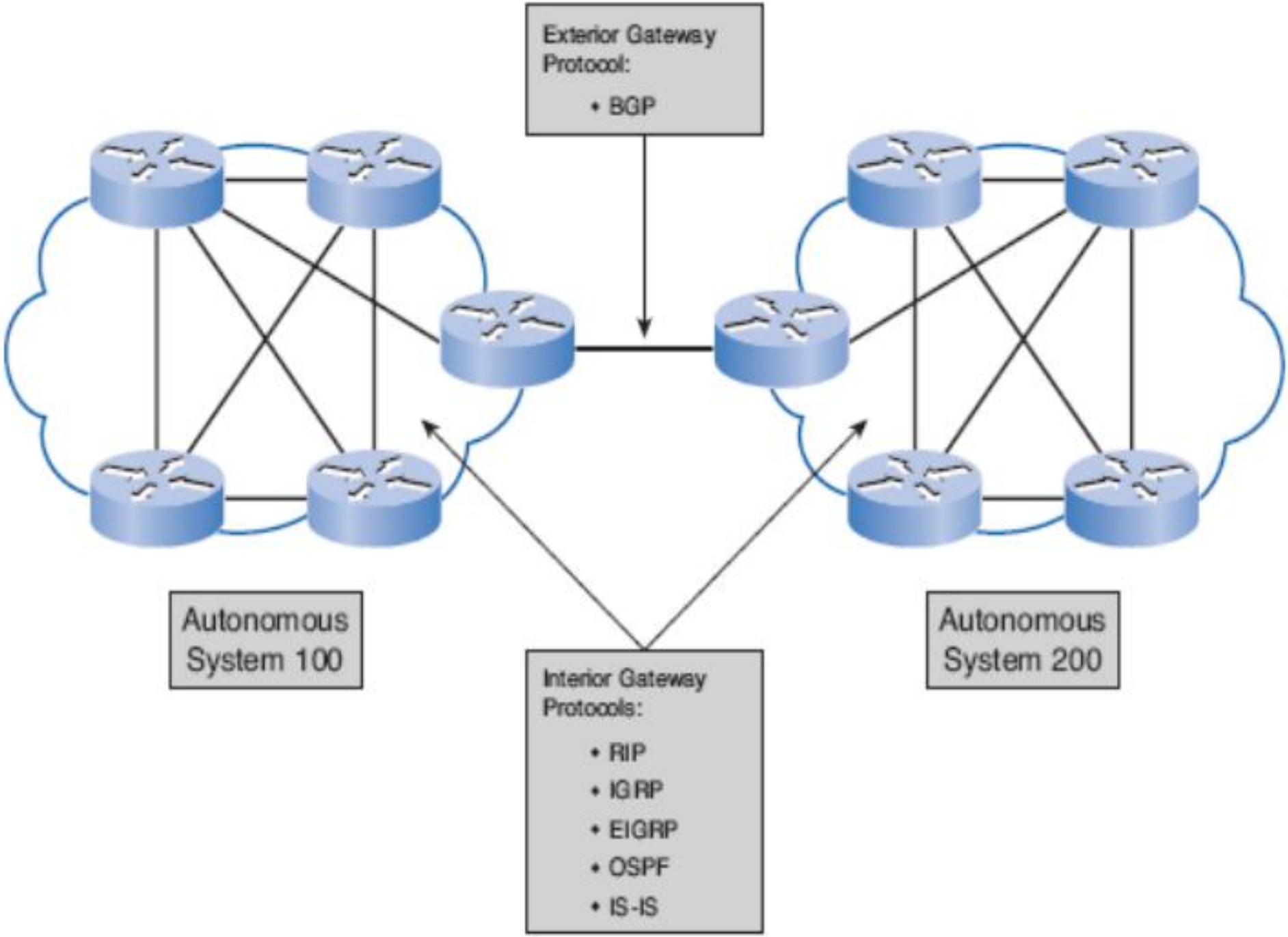
- **Dynamic routing advantages:**
  - Administrator has less work maintaining the configuration when adding or deleting networks.
  - Protocols automatically react to the topology changes.
  - Configuration is less error-prone.
  - More scalable, growing the network usually does not present a problem.
- **Dynamic routing disadvantages:**
  - Router resources are used (CPU cycles, memory and link bandwidth).
  - More administrator knowledge is required for configuration, verification, and troubleshooting.

Dynamic Routing Protocols Classification

- Routing protocols can be classified into different groups according to their characteristics.
- The most commonly used routing protocols are:
  - **RIP** - A distance vector **interior** routing protocol
  - **IGRP** - The distance vector **interior** routing protocol
  - **OSPF** - A link-state **interio**r routing protocol
  - **IS-IS** - A link-state **interior** routing protocol
  - **EIGRP** - The advanced distance vector **interior** routing protocol
  - **BGP** - A path vector **exterior** routing protocol

IGP and EGP

- **An autonomous system (AS**) - otherwise known as a **routing domain** - is a **collection of routers under a common administration.**
- Typical examples are a company's internal network and an Internet service provider's network.
- Because the Internet is based on the autonomous system concept, two types of routing protocols are required: **interior and exterior** routing protocols.
- These protocols are:
  - **Interior Gateway Protocols (IGP)** are used for **intra-autonomous system routing -** routing **inside** an autonomous system.
  - **Exterior Gateway Protocols (EGP)** are used for **inter-autonomous system routing –** routing **between** autonomous systems.


Exterior Gateway Protocol:
• BGP
Autonomous System 100
Autonomous System 200
Interior Gateway Protocols:
• RIP
• IGRP
• EIGRP
• OSPF
• IS-IS

Distance Vector Routing Protocol Operation

- Distance vector means that routes are advertised as vectors of distance and direction.
- Distance is defined in terms of a metric such as hop count and direction is simply the next-hop router or exit interface.
- Distance vector protocols typically use the **Bellman-Ford** algorithm for the best path route determination.
- Some distance vector protocols periodically **send complete routing tables to all connected neighbors.**
- In large networks, these routing updates can become enormous, causing significant traffic on the links.
- Although the Bellman-Ford algorithm eventually accumulates enough knowledge to maintain a database of reachable networks, the algorithm **does not allow** a router to know the exact topology of an internetwork.

Distance Vector Routing Protocol Operation

- The router only knows the routing information received from its neighbors.
- The only information a router knows about a remote network is the distance or metric to reach that network and which path or interface to use to get there.
- These protocols **do not have an actual map of the network topology.**
- Distance vector protocols work best in situations **where**:
  - The network is **simple and flat,** and does not require a special hierarchical design.
  - The administrators do not have enough knowledge to configure and troubleshoot link-state protocols.
  - Specific types of networks, such as hub-and-spoke networks, are being implemented.
  - **Worst-case convergence times** in a network are **not a concern**.

Link-state Protocol Operation

- In contrast to distance vector routing protocol operation, a router configured with a link-state routing protocol can create a **"complete view" or topology of the network by gathering information from all of the other routers.**
- A link-state router uses the **link-state information** to create a **topology map** and to select the best path to all destination networks in the topology.
- With some distance vector routing protocols, routers send **periodic updates** of their routing information to their neighbors.
- Link-state routing protocols **do not use periodic updates.**
- **After the network has converged,** a link-state update **only sent when there is a change in the topology.**
- Link-state protocols work best in situations where:
  - The network design is hierarchical, usually occurring in **large** networks.
  - The administrators have a good knowledge of the implemented link-state routing protocol.
  - **Fast convergence** of the network is **crucial.**

Administrative Distances

- Routers learn about adjacent networks that are directly connected and about remote networks by using static and dynamic routing protocols.
- **In fact, a router might learn of a route to the same network from more than one source.**
- Although less common, more than one dynamic routing protocol can be deployed in the same network.
- In some situations it may be necessary to route the same network address using multiple routing protocols such as RIP and OSPF.
- **So, how does a router determine which route to install in the routing table when it has learned about the same network from more than one routing source?**
- **Administrative distance (AD) defines the preference of a routing source.**
- Each routing source - including specific routing protocols, static routes, and even directly connected networks - is **prioritized** in order of most- to least-preferable using an **administrative distance value.**

Administrative Distances, cont…

- Administrative distance is an integer value from **0 to 255**.
- The **lower** the value the **more preferred** the route source.
- An administrative distance of 0 is the most preferred.
- Only a directly connected network has an administrative distance of 0, which cannot be changed.

The AD value can be verified by the Show IP route Command.

The AD value can also be verified with the **show ip protocols** command.

Administrative distance

Route source	Administrative distance
Connected	0
Static	1
EIGRP summary route	5
External BGP	20
Internal EIGRP	90
IGRP	100
OSPF	110
IS-IS	115
RIP	120
External EIGRP	170
Internal BGP	200